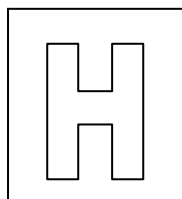


Candidate Name: _____

Class Adm No

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2021 End-of-Year Exams Pre-University 2

BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

15 September 2021

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 26
2	/ 10
3	/ 9
Section B	/ 15
Total	/ 60

This question paper consists of 24 printed pages, including 3 blank pages.

[Turn over

Section A [Total: 45 marks]

Answer **all** questions in this section.

1. Proteins are one of the biomolecules of life that play various roles in a cell.

- (a) State two other biomolecules that can be found in cells.

Nucleic acids, lipids, carbohydrates (A! any named examples)
; any 2 for 1 mark

.....[1]

- (b) Enzymes are a common example of proteins found in cells. Enzymes are known to denature at temperatures beyond their optimum temperature.

- (i) Explain how high temperature causes enzymes to denature.

1. High temperatures results in increase in kinetic energy;
2. Which breaks weak R-group interactions/name any 2(hydrophobic interactions, hydrogen bonds, ionic bonds); (compulsory)
3. Polypeptide chain unfolds;
4. Loss of 3D conformation/globular structure;
; for 1 mark, max 1

.....[3]

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At high temperatures beyond optimum temperatures, enzymes show decreased to no activity due to denaturation.

At 0°C, most enzymes tends to show relatively low to no activity at all. However, it is not considered denatured, but rather **inactive**.

At sub-zero temperatures (below 0°C), enzyme activity is completely stopped due to '**cold denaturation**'.

A group of researchers used computer simulation to investigate how protein molecules like enzymes behaved in an aqueous environment at three different temperatures:

- sub-zero
- optimum
- beyond optimum

Fig. 1.1 shows the results.

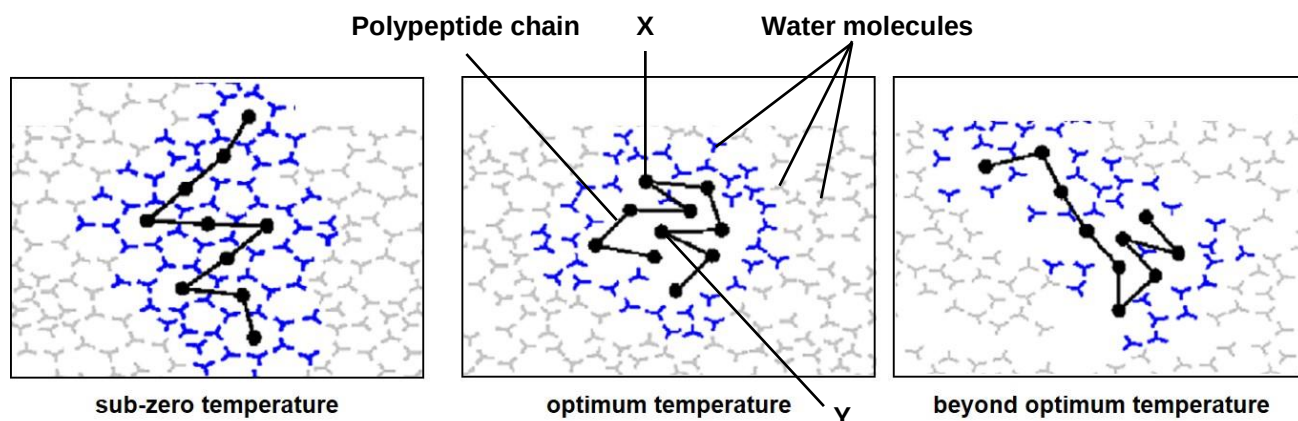


Fig. 1.1

X and Y represents amino acids in a polypeptide chain.

(ii) State the type of R-groups amino acids X and Y would likely to have.

X: .. X: Hydrophilic/charged/polar;
Y: .. Y: Hydrophobic/non-polar R:uncharged
; for 1 mark

[2]

(iii) With reference to the information given and Fig. 1.1, compare the **structure** and **activity** of the enzyme when it is considered:

- 'inactive' and
- after undergoing 'cold denaturation'

Structure

1. R-group interactions are disrupted/broken in cold denaturation, whereas R-group interactions are intact/not broken for enzyme being inactive at 0°C;
2. Loss of 3D conformation / unfolding of polypeptide chain for cold denaturation at sub-zero temperatures compared to no loss of 3D conformation / no unfolding when enzyme is inactive at 0°C;
3. Hydrophobic regions are exposed to the exterior when enzymes undergo cold denaturation, whereas hydrophobic regions mains embedded in the core of the enzyme when it is inactive;
4. Cold denaturation is irreversible, whereas inactive is reversible/temporary;
; for 1 mark, max 2

Activity

5. Both will have no enzyme activity/0 rate of reaction;
; for 1 mark

[3]

Freezing temperatures can also damage the cell surface membrane and membranes within the cell.

(c) (i) State an example of membranes found within the cell.

Any membrane bound organelles;
; for 1 mark

[1]

(ii) With reference to the structure and properties of membranes, explain why the transport of large molecules would no longer be possible at freezing temperatures.

Membrane is not of optimal fluidity /not fluid enough/OWTTE;
Vesicles would not be able to form / invagination of membrane would not be able to occur;
; for 1 mark

[2]

A method called *in vitro* translation is often used by scientists to produce proteins in laboratory. The proteins produced can then be purified and isolated for use in subsequent experiments. This method uses extracts from animal cells, plant cells or bacteria. The method involves the following steps:

- select for cells that have high levels of protein synthesis.
- cells are treated so that cell walls, if present and cell membranes are broken down.
- cell extract is then treated to destroy cell's own DNA and mRNA.
- desired mRNA from any source is then added to these extracts, it will be translated into the corresponding protein.

(d) Suggest a reason why:

(i) the cell's own mRNA needs to be destroyed.

1. only the desired protein is produced/OWTTE; ORA
2. unwanted protein does not have to be separated from desired protein;
3. idea that inefficient process if translation machinery used to produce other proteins;
4. cell's proteins may, inhibit/affect/ hinder/OWTTE, process
; for 1 mark, max 1

.....
.....[1]

(ii) mRNA from any source can be translated in any type of extract.

1. Universal genetic code;
2. All cells use mRNA as template for translation;
3. Ref to ribosome function/translation is the same in all cells;
; for 1 mark, max 2

.....
.....
.....
.....[2]

(e) Describe two differences between the cell structures used in translation in prokaryotes and eukaryotes.

1. Prokaryotes uses 70S ribosomes whereas eukaryotes uses 80S ribosomes;
2. Prokaryotes only have free ribosomes/only in cytoplasm whereas eukaryotes have free/cytoplasm and bound ribosomes on RER;
3. Ref to different rRNA (no need to specify);
; for 1 mark, max 2

.....[2]

Proteins play a variety of key roles in cellular functions. Defective proteins due to mutations can affect various cellular processes and lead to diseases like cancer.

Mutations can be classified as gene mutation or chromosome aberration.

(f) (i) Fill in Table 1.1 to identify:

- One mechanism/cause of gene mutation and chromosome aberration
- The **corresponding effect** of the mechanism/cause of the gene mutation and chromosome aberration

Table 1.1

	mechanism/cause	corresponding effect
gene mutation	<u>Nucleotide</u> substitution/deletion/ addition/inversion; ; for 1 mark	Substitution: Missense/silent/nonsense Deletion/Addition: frameshift/nonsense/longer/ shorter polypeptide Inversion: missense/nonsense ; for 1 mark
chromosome aberration	Translocation/duplication/ deletion/inversion; OR Nondisjunction ; for 1 mark	Translocation: change in level of gene expression Duplication: additional copies of genes Deletion: loss of genes Inversion: change in level of gene expression OR Nondisjunction: aneuploidy/polyploidy : for 1 mark

[4]

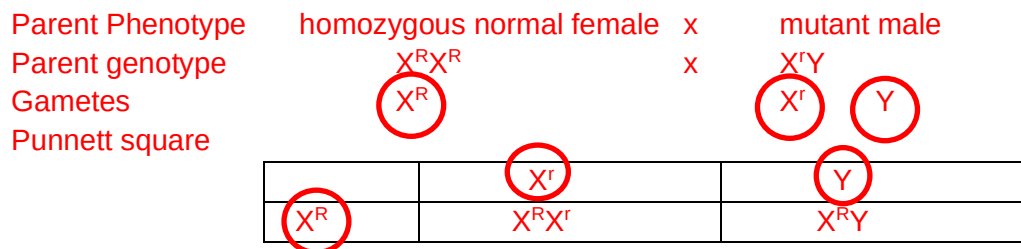
Recent research has shown that a gene on the X-chromosome may contribute to ovarian cancer predisposition. Having one copy of the mutant allele increases the risk of developing ovarian cancer, while having two copies of the mutant allele further increases the risk of developing ovarian cancer.

- (ii) Using your knowledge of cancer, explain why a female homozygous for the mutant allele does not necessarily develop cancer.

1. Cancer development is a multi-step process/require multiple mutations;
2. Require mutations to a number of tumour suppressor genes and proto-oncogenes;
3. Mutations are random, and there is a chance that the female will not acquire the necessary mutations/OWTTE

[2]

- (ii) Using the symbols X^R to represent the normal allele and X^r to represent the mutant allele, draw a genetic diagram to show how a female homozygous for the normal allele will have all daughters being heterozygous for the mutant allele.



F1 Genotype $X^R X^r$: $X^R Y$

F1 Phenotype heterozygous female : normal male

F1 Phenotypic ratio 1 : 1

1. correct parent genotype;
2. correct gametes formed (allow e.c.f);
3. correct F1 phenotypic ratio;

[3]

[Total: 27]

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2. The hedgehog, *Erinaceus europaeus*, is a small carnivorous mammal native to Northern Europe.

Hedgehogs were introduced onto a small group of islands off the west coast of Scotland in 1974. The hedgehog population has increased so that there are now over 5 000 breeding pairs. These hedgehogs have no natural predators on these islands and their diet consists mainly of bird's eggs.

Fig. 2.1 shows the distribution of *Erinaceus europaeus* population across the north and south island off the west coast of Scotland in the year 2000.

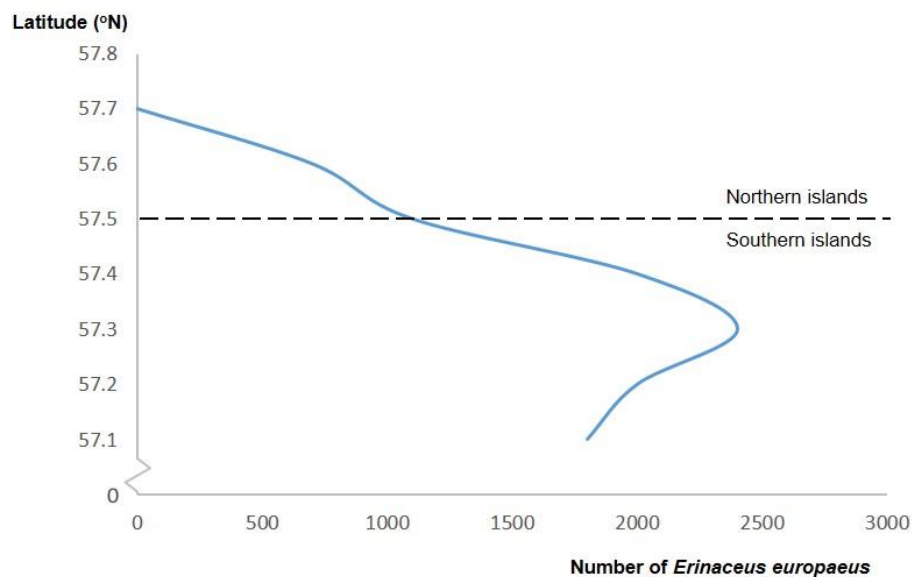


Fig. 2.1

Table 2.1 shows the changes in the population of the species of birds found on the same islands from 1983 to 2000.

Table 2.1

	breeding pairs in 1983	breeding pairs in 2000	% change in population
north island			
oystercatcher	928	1122	+21
lapwing	1104	1364	+24
redshank	486	733	+51
south island			
oystercatcher	907	1403	+55
lapwing	1869	1287	-31
redshank	1288	760	-41

- (a) (i) Calculate the % change in population of oystercatcher in the north and south island respectively. **Write your answers in Table. 2.1**, leave your answers to the **nearest percentage**.

Both correct for 1 mark;

[1]

- (ii) Using Fig. 2.1 and Table. 2.1, account for the difference in % change in population of **redshank**.

North island

1. Fewer/less hedgehogs, redshank eggs are not hunted as often/able to develop into adult;
2. Increase from 486 breeding pairs in 1983 to 733 in 2000 / 51% increase

South island

3. presence of hedgehogs/more hedgehogs causes more redshank eggs to be hunted, less survive to adult;
4. decrease from 1288 breeding pairs in 1983 to 760 in 2000/41% decrease;
; for 1 mark

[4]

- (iii) Suggest reasons for the increase in oystercatcher population on the south island.

1. Better camouflaged / distasteful / inaccessible
2. Hedgehogs preference for lapwing and redshank, doesn't hunt for oystercatcher eggs
3. Oystercatchers have less competition, able to survive better and more are able to breed and lay more eggs;
4. AVP;
; for 1 mark, max 2

[2]

Variation plays an important role in the process of evolution by natural selection.

Gene mutation is one source of variation.

(b) Apart from gene mutation, outline one other source of variation and explain why variation is important in natural selection.

Source of variation:

crossing over during prophase I / independent assortment / random fertilisation;
; for 1 mark

Why variation is important:

1. Increase gene pool / having a range of phenotypes increase the probability that one of the phenotype will be at selective advantage / selected for, that can survive and reproduce to pass on favourable alleles;
OR
2. Not all phenotypes will be at selective disadvantage / selected against;
; for 1 mark
3. Ensure survival of species/prevent extinction;
; for 1 mark

[3]

[Total: 9]

3. Maize, *Zea mays*, is a major cereal food crop.

The expected increase in carbon dioxide concentration in the atmosphere is expected to increase the rate of growth of many crop plants.

(a) (i) State the exact location in a plant cell where carbon dioxide is used.

.. Stroma;
; for 1 mark[1]

Carbon dioxide is a gas.

(ii) Outline how plants use carbon dioxide for growth.

1. <u>Calvin cycle /light independent stage/ dark reaction;</u> 2. Carbon dioxide is fixed with <u>RuBP</u> , catalysed by <u>Rubisco</u> to form PGA; 3. PGA is reduced to <u>G3P</u> by ATP and NADPH; 4. Ref to every 3 carbon dioxide fixed, one G3P is produced, while 5 other G3P are <u>regenerated to from RuBP</u> ; ; for 1 mark, max 3 5. Ref to G3P can be <u>converted into other biomolecules</u> for growth (or give example: starch, cellulose) 6. Ref to G3P can be converted into <u>glucose for respiration</u> to release ATP for growth; ; for 1 mark, max 1	
--	--

.....[4]

Investigations were carried out into the effect of increased carbon dioxide concentration on the rate of photosynthesis in maize.

- Maize plants were grown in open-air trials, in the same field and were exposed to the same changes in the weather (with the exception of carbon dioxide concentrations).
- 50% of the plants were exposed to a normal carbon dioxide concentration.
- 50% of the plants were exposed to an increased carbon dioxide concentration.
- The rate of photosynthesis was measured as the net assimilation rate of carbon dioxide which indicates the overall amount of carbon dioxide taken in by the plant.
- Measurements were made at three-hourly intervals between 0700 hours and 1900 hours on three different days.

The results are shown in Fig. 3.1.

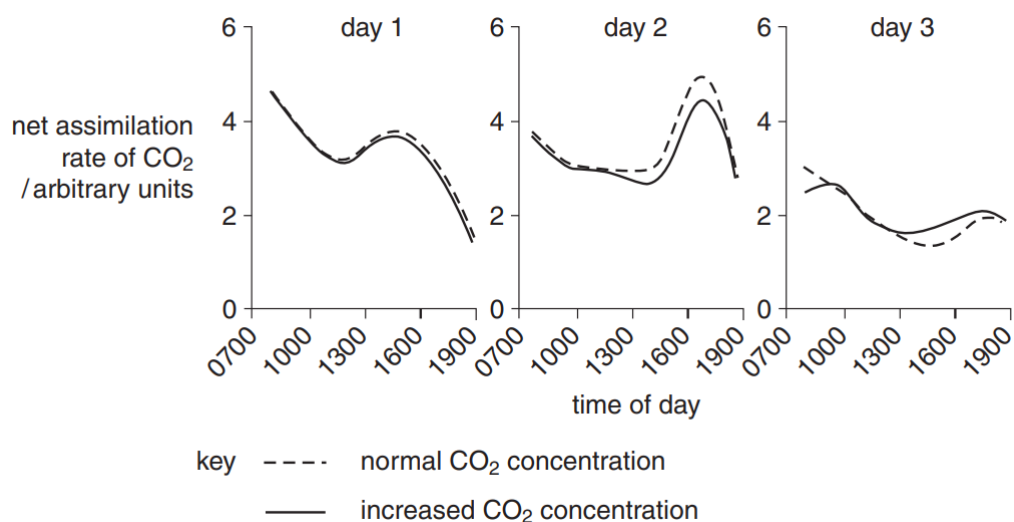


Fig. 3.1

(b) Account if the results shown in Fig. 3.1 supported the expected increase in rate of growth of crop plants in increased CO₂ concentration.

1. No;
; for 1 mark
 2. For 2 out of 3 days, increased CO₂ concentration actually showed an overall lower net assimilation rate of CO₂ as compared to normal CO₂ concentration;
 3. Only day 3, 1300 to 1800 showed an increase net assimilation rate of CO₂ under increased CO₂ concentration as compared to normal CO₂ concentration;
- [2]
; for 1 mark, max 1

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Plants under heat stress are exposed to high temperatures beyond a threshold level for a period of time that results in permanent damage to the plant's growth and development. While plants under dry stress lose excessive amount of water.

Both heat stress and dry stress can be attributed to climate change and one of the main drivers of climate change is the rise in atmospheric carbon dioxide levels.

Fig. 3.2 shows the predicted intensity of heat stress in Africa and dry stress in South America.

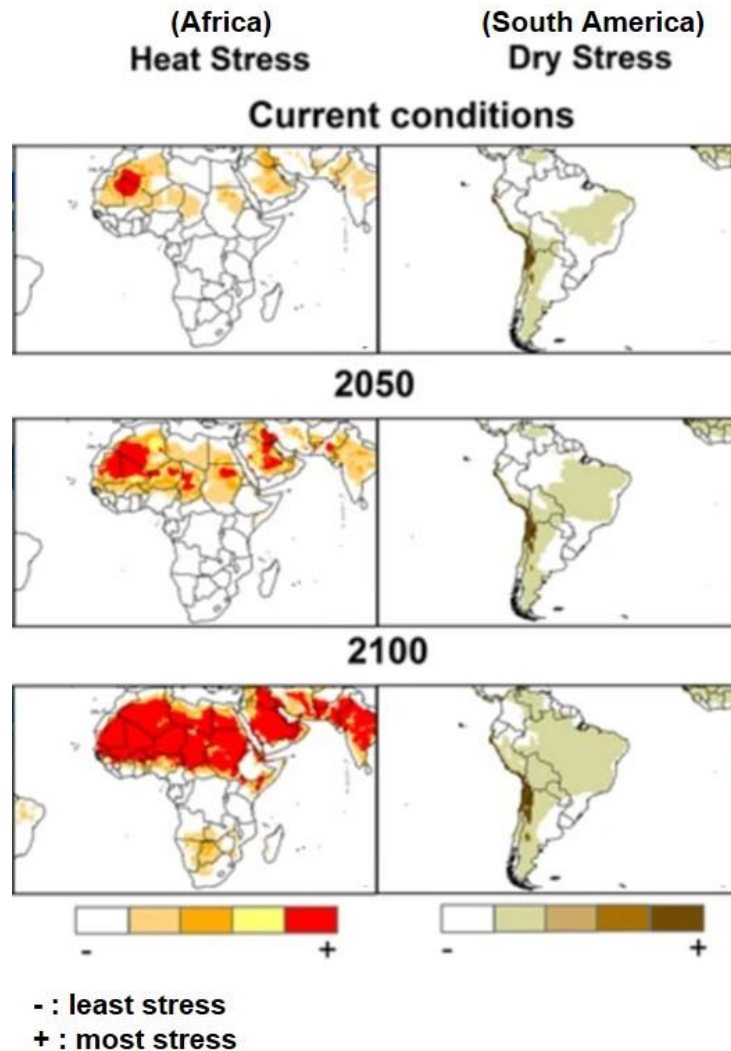


Fig. 3.2

The current distribution of maize crops are shown in Fig. 3.3. Computer modelling was used to predict the changes to distribution of maize crops in 2050 and 2100. Conditions are ranked from most optimal for growth to unsuitable for growth in the following order: Optimal > Suitable > Marginal > Unsuitable.

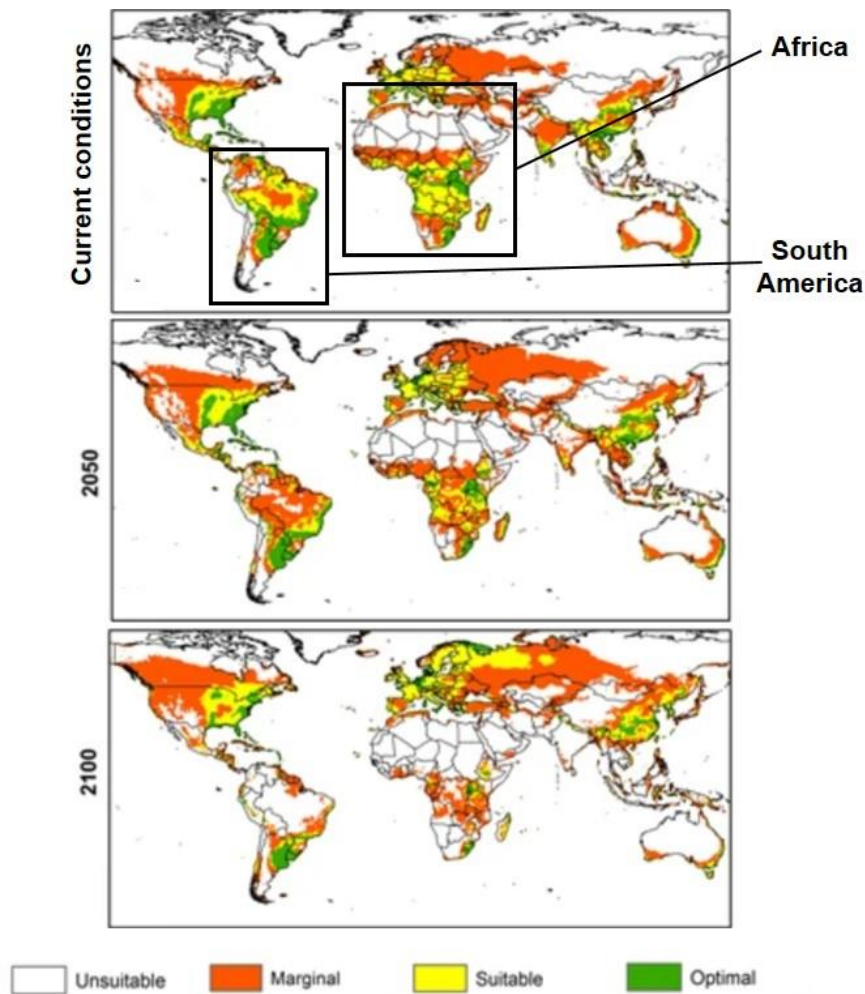


Fig. 3.3

(c) With reference to Fig. 3.2 and Fig. 3.3, account for the change in distribution of maize crops in South America.

1. Largely suitable in current conditions, but northern region gradually becomes marginal and unsuitable in 2100/OWTTE;
 2. Due to increase dry stress in the northern region;
- ; for 1 mark

2]

[Total: 9]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4

- (a)** Describe the roles played by proteins in mitotic cell cycle.

[5]

- (b)** With reference to named examples, explain the importance of the structure of biopolymers to their functions.

[10]

[Total: 15]

5

- (a)** Describe the roles played by proteins in cellular respiration.

[5]

- (b)** Explain how mutations in stem cells can affect an individual and also subsequent generations.

[10]

[Total: 15]

End of Paper

4.

(a) Describe the roles played by proteins in mitotic cell cycle.

[5]

1. Ref to tumour suppressor proteins involved in detecting DNA damage/cell cycle arrest/DNA repair/apoptosis (any one function);
2. Ref to proteins coded by proto-oncogenes involved in promoting cell progression in cell cycle / stimulate growth / prevent apoptosis;
3. Ref to cyclins as part of cell cycle check points to activate cyclin-dependent kinase;
4. Ref to cyclin-dependent kinase that regulates the cell cycle checkpoint;
5. Ref to any proteins involved in DNA replication:
 - a. Helicase – unwind DNA;
 - b. Topoisomerase – release strain in DNA helix
 - c. Single stranded DNA binding proteins – stabilise replication fork;
 - d. Primase – synthesise RNA primer;
 - e. DNA polymerase III – adds complementary base pairs to elongate daughter strand/catalyse formation of phosphodiester bond between adjacent nucleotides;
 - f. DNA polymerase I – removes RNA primer and replace with DNA nucleotides;
 - g. Ligase – seals nicks between DNA fragments by catalysing formation of phosphodiester bond;
; max 2
6. Kinetochore proteins binding to centromere region to allow binding of spindle fibres
7. Spindle fibres lengthen and shorten to attach and pull apart sister chromatids
8. AVP
; for 1 mark

- (b) With reference to named examples, explain the importance of the structure of biopolymers to their functions.

[10]

	Structure	Function
	Carbohydrates	
	(A) Starch/Glycogen	
1	Starch is a <u>large</u> polysaccharide which makes it <u>insoluble</u> in water;	a <u>good energy storage</u> because large amount of glucose can be stored <u>without affecting the water potential</u> of cell;
2	<u>coiled into helices</u> ;	folded in a <u>compact</u> shape and hence <u>large amount of starch can be stored</u> within a cell;
3	made up of <u>large number of glucose monomers</u> linked by glycosidic bonds;	Upon hydrolysis of glycosidic bonds, <u>large amount of glucose are released for respiration</u> , releasing energy for cellular functions;
4	consists of α -glucose monomers linked by α -(1,4)- <u>glycosidic bonds</u> and further branched by α -(1,6)-glycosidic bonds;	The <u>glycosidic linkages in starch can be hydrolysed easily</u> into large number of monomers, acting as <u>an accessible store of glucose</u> ;
5	<u>highly branched</u> with multiple free ends due to presence of $\square$ <u>α-(1,6)-glycosidic bonds</u> ;	The highly branched amylopectin is very <u>efficient for fast generation of glucose</u> because cleavage takes place at the <u>multitude of free ends simultaneously</u> ;
	(B) Cellulose	
1	Cellulose is a <u>large</u> polysaccharide which makes it <u>insoluble</u> in water; OR There are <u>no free -OH group to form hydrogen bonds with water</u> as they are all used in forming cross-linkage;	Cellulose insolubility ensures that the <u>structural support does not dissolve</u> away when exposed to water;
2	The <u>successive beta-glucose units are rotated 180° to each other</u> , thus -OH group projects out in both direction to form <u>hydrogen bonds with neighbouring cellulose chains/hydrogen bond cross-links</u> ;	Cellulose has <u>high tensile strength</u> and rigidity; OR Cellulose is <u>resistant to stretching</u> and thereby prevents plant cells from collapsing; OR
3	Cellulose chains <u>bundle</u> together to form <u>microfibrils</u> and in turn form <u>macrofibrils</u> ;	Rigidity <u>prevents cells from bursting</u> when water enters cells by osmosis;

3	Cellulose has <u>large intermolecular spaces</u> between microfibrils;	Cellulose is thus <u>fully permeable</u> , allowing the passage of water and solutes across cellulose cell wall in plants;
Proteins		
(C) Haemoglobin		
1	Each subunit of haemoglobin folds and coils into <u>compact</u> globular shape.	Compact structure allows <u>more packing of haemoglobin into red blood cells</u> for more efficient transport of oxygen.
2	Each of the 4 <u>subunit</u> carries a <u>prosthetic haem group</u> with iron.	The <u>capacity to carry 4 haem group</u> increase efficiency of oxygen transport;
3	The iron is capable of <u>reversibly binding</u> one <u>oxygen molecule</u> each;	Reversible binding of oxygen allows for <u>uptake and release of oxygen</u> ;
4	The binding of one haem group will cause a <u>conformational change</u> in the remaining subunits, making it <u>easier for the binding of oxygen</u> to other haem groups;	<u>Cooperative binding</u> facilitates <u>easier binding and release of oxygen</u> ;
5	Each polypeptide chain in haemoglobin consists of both hydrophilic and hydrophobic amino acid residues. The secondary structure of the polypeptide is folded such that the bulk of the <u>hydrophobic amino acid residues</u> are buried in the <u>interior</u> of the globular structure while the <u>hydrophilic amino acid residues</u> are on the <u>outside</u> ;	Haemoglobin is <u>soluble</u> in <u>aqueous environment of red blood cells</u> to <u>bind to oxygen</u> that has been <u>dissolved</u> in the blood;
6	The structure of globin is globular except for a <u>deep hydrophobic cleft</u> that consists of the haem binding site;	The haem binding site provides a <u>hydrophobic environment for the binding of haem</u> which is largely hydrophobic;
Nucleic acid		
(D) DNA		
1	DNA contains a <u>large amount of nitrogenous bases</u> ;	Capable of carrying <u>large amount of coded information</u> within the cell;
2	DNA is highly packed and condensed	Allow DNA to be more <u>compact</u> and large amount can be stored in the nucleus.
3	extensive number of <u>hydrogen bonds between complementary nitrogenous bases</u> along opposite strands.help stabilize the structure of the double helix; <u>Hydrophobic interactions between stacked nitrogenous bases</u> further contribute to the stability of the DNA helix structure;	<u>Stable</u> structure so that <u>mutations occur rarely</u>

	Covalent <u>phosphodiester bonds linking adjacent nucleotides</u> give rise to a strong phosphodiester backbone that help maintain the integrity of the DNA base sequence.; The <u>lack of hydroxyl (–OH) groups at 2' carbon</u> of deoxyribose also reduces the reactivity of DNA to increase its stability; ; max 2	
4	<u>Double stranded, each strand</u> is able to act as <u>template</u> for synthesis of daughter strand via <u>complementary base pairing</u> ;	Able to <u>replicate accurately</u> via semi-conservative model;
5	<u>Double stranded, intact strand</u> can be used as <u>template</u> to guide repair via <u>complementary base pairing</u> ;	Able to <u>repair damage</u> ;
6	<u>Template strand</u> allows for <u>synthesis of mRNA</u> via <u>complementary base pairing</u> ;	Able to <u>pass on genetic information to mRNA</u> via transcription for protein synthesis;
(E) RNA (m/r/tRNA counts as 1 example)		
1	mRNA is <u>single stranded</u> ;	Allows <u>ribosome</u> to read it as <u>template</u> for translation;
2	mRNA has <u>5'-cap</u>	Allows <u>small ribosomal subunit</u> to <u>recognise and bind</u>; OR <u>Facilitate the export of mRNA out of nucleus</u>; OR <u>Protects degradation of mRNA</u> from RNAses/nucleases/enzymes in cytosol
3	mRNA has <u>5'UTR /3'UTR</u>	Facilitate <u>binding of ribosomes</u> ;
4	mRNA contains <u>codons</u>	Allow the <u>binding of specific tRNA with correct amino acid</u> ;
5	mRNA has <u>poly-A-tail</u> ;	<u>Delay degradation of mRNA</u> by RNAses/nucleases/enzymes in cytosol;
6	rRNA folds back on itself to form a specific 3D conformation; ; max 2	Allows formation of <u>active site of peptidyl transferase</u> to <u>catalyse formation of peptide bond</u> in large ribosomal subunit; Allows rRNA <u>to assemble with ribosomal proteins</u> to form ribosomal subunits; Facilitate <u>binding of mRNA condons with tRNA anti-codons</u>; Facilitate <u>binding of small ribosomal subunit with mRNA</u>;
7	tRNA folds back on itself to form a specific <u>clover-leaf shape</u> ;	Allows it to <u>bind to EPA sites of ribosomes</u>; OR Allows it to <u>bind to active site of amino-acyl tRNA synthetase</u>;

	tRNA has a <u>3' CCA end</u>	Allows for <u>attachment of specific amino acid</u> via ester bond;
	tRNA has an <u>anti-codon</u> region	Allows for <u>complementary base pairing with codon on mRNA</u> to ensure the <u>correct amino acid</u> is joined to form the polypeptide chain;

QWC: provide at least 2 named examples

[Total: 15]

5.

(a) Describe the roles played by proteins in cellular respiration.

[5]

1. Ref to series of enzymes (no need to specify) involved in glycolysis to break down/oxidise glucose to pyruvate;
 2. Ref to enzyme pyruvate dehydrogenase complex convert pyruvate to acetyl-CoA;
 3. Ref to enzymes involved in Krebs cycle;
 4. Ref to enzymes involved in oxidative decarboxylation;
 5. Electron transport chain complexes involved in transport of electron;
 6. Energy released allows the complexes to pump H⁺ from matrix to intermembrane space;
 7. ATP synthase provides a hydrophilic channel for diffusion of H⁺ from intermembrane space to matrix;
 8. ATP synthase uses the energy released from diffusion of H⁺ to synthesise ATP from ADP and Pi;
 9. Ref to enzyme lactate dehydrogenase/ethanol dehydrogenase convert pyruvate to lactate/ethanol during anaerobic respiration;
- ; for 1 mark, max 5

- (b) Explain how mutations in stem cells can affect an individual and also subsequent generations.

[10]

Functions of stem cells

1. Stem cells function for the growth and development of an organism;
2. Stem cells function for the replacement of worn-out/damaged cells;
; for 1 mark

Mechanisms of mutations

3. Ref to mechanisms of gene mutations: substitution, deletion, insertion and inversion; (any 2)
4. Ref to mechanisms of chromosomal structural mutations: deletion, duplication, inversion and translocation; (any 2)
5. Ref to mechanisms of chromosomal numerical mutations: non-disjunction of sister chromatids during anaphase of mitosis / anaphase II of meiosis II OR homologous chromosomes during anaphase I of meiosis I;
; for 1 mark, max 2

Effect of mutations

6. Ref to missense, nonsense, frameshift mutations, silent; (any 2);
7. Ref to gain/loss of genes (from chromosomal structural mutation);
8. Ref to aneuploidy and polyploidy;
; for 1 mark, max 2

Within an individual

9. Ref to somatic mutations only affect individual;
10. Stem cells divide by mitosis;
11. Any mutations acquired by (adult) stem cells if undetected will be passed on to daughter cells;
12. If mutations occur to genes coding for important protein product involved in the function of organs the function of the organ may be affected/OWTTE;
13. If mutations occur to proto-oncogenes/tumour suppressor genes, it may lead to increase risk in cancer development;
14. The individual will have a mix of cells with mutations that derived from a stem cell with the mutation and normal cells that derived from stem cells without mutations/OWTTE;
; for 1 mark, max 2

Subsequent generations

15. If mutation occurs to stem cells forming germ cells, the gametes formed will contain the mutation but the individual will not be affected/germline mutation;
16. Upon fertilisation, the zygote formed will contain the mutation;
17. Zygote is a totipotent stem cell that divides to form the entire organism;
18. As such , all cells of individual in the next generation will contain the mutation;
19. Ref to mutations coding for favourable/unfavorable traits will be selected for/against;
20. Ref to change in allele frequency;
; for 1 mark, max 2

QWC: Student shows an understanding that non-germline mutations will only affect individual and not the next generation